

Question-7.4.29

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Question

If one of the diameters of the circle $x^2 + y^2 - 2x - 6y + 6 = 0$ is a chord to the circle with the center $(2, 1)$, then the radius of the circle is

- ① $\sqrt{3}$
- ② $\sqrt{2}$
- ③ 3
- ④ 2

Solution

Given the circle equation:

$$x^2 + y^2 - 2x - 6y + 6 = 0 \quad (1)$$

$$\mathbf{x}^T \mathbf{x} + 2\mathbf{u}^T \mathbf{x} + f = 0, \text{ where } \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} g \\ f \end{pmatrix} \quad (2)$$

From the given equation:

$$2\mathbf{u}^T \mathbf{x} = -2x - 6y \implies \mathbf{u} = \begin{pmatrix} -1 \\ -3 \end{pmatrix}, \quad f = 6 \quad (3)$$

Center of the circle is:

$$\mathbf{c}_1 = -\mathbf{u} = \begin{pmatrix} 1 \\ 3 \end{pmatrix} \quad (4)$$

Solution

Radius of the circle is:

$$r_1 = \sqrt{\mathbf{u}^T \mathbf{u} - f} = \sqrt{(-1)^2 + (-3)^2 - 6} = \sqrt{4} = 2 \quad (5)$$

Given the second circle has center:

$$\mathbf{c}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad (6)$$

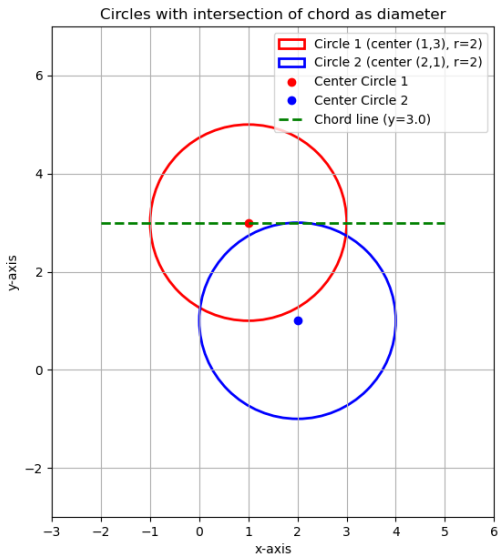
Assuming the diameter chord line of the first circle with equation $y = 3$,

$$\mathbf{n} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad c = 3 \quad (7)$$

Distance from second circle center $\mathbf{c}_2$ to this chord:

$$\frac{|\mathbf{n}^T \mathbf{c}_2 - c|}{\|\mathbf{n}\|} = \frac{|1 - 3|}{1} = 2 \quad (8)$$

Therefore, the radius of the second circle $r=2$.



```
// code.c
#include <stdio.h>

void get_circle_data(double *c1, double *r1, double *c2, double *
    r2, double *chord_y) {
    // Circle 1 center (1,3), radius 2
    c1[0] = 1; c1[1] = 3;
    *r1 = 2;

    // Circle 2 center (2,1), unknown radius
    c2[0] = 2; c2[1] = 1;
    *r2 = 2; // radius determined by chord distance

    // chord line y=3
    *chord_y = 3;
}
```

```
# Code by GVV Sharma  
# Modified for Problem Solution  
# Released under GNU GPL  
# Calculating area enclosed between curves  
# call.py  
import ctypes  
import numpy as np  
  
lib = ctypes.CDLL('./code.so')
```

```
get_circle_data = lib.get_circle_data
get_circle_data.argtypes = [
    np.ctypeslib.ndpointer(dtype=np.float64, shape=(2,)),
    ctypes.POINTER(ctypes.c_double),
    np.ctypeslib.ndpointer(dtype=np.float64, shape=(2,)),
    ctypes.POINTER(ctypes.c_double),
    ctypes.POINTER(ctypes.c_double)
]
get_circle_data.restype = None
def get_data():
    c1 = np.zeros(2, dtype=np.float64)
    r1 = ctypes.c_double()
    c2 = np.zeros(2, dtype=np.float64)
    r2 = ctypes.c_double()
    chord_y = ctypes.c_double()

    get_circle_data(c1, ctypes.byref(r1), c2, ctypes.byref(r2),
                    ctypes.byref(chord_y))
    return c1, r1.value, c2, r2.value, chord_y.value
```


Python-Code

```
import matplotlib.pyplot as plt
import numpy as np
from call import get_data

c1, r1, c2, r2, chord_y = get_data()

fig, ax = plt.subplots(figsize=(7,7))

# Full circle 1 (center c1, radius r1)
circle1 = plt.Circle(c1, r1, color='r', fill=False, linewidth=2,
    label='Circle 1 (center (1,3), r=2)')

# Full circle 2 (center c2, radius r2)
circle2 = plt.Circle(c2, r2, color='b', fill=False, linewidth=2,
    label='Circle 2 (center (2,1), r=2)')

ax.add_artist(circle1)
ax.add_artist(circle2)
```

```
# Plot centers
ax.plot(c1[0], c1[1], 'ro', label='Center Circle 1')
ax.plot(c2[0], c2[1], 'bo', label='Center Circle 2')
# Plot chord line y = chord_y
x_vals = np.linspace(min(c1[0], c2[0]) - 3, max(c1[0], c2[0]) +
                      3, 400)
y_vals = np.full_like(x_vals, chord_y)
ax.plot(x_vals, y_vals, 'g--', linewidth=2, label=f'Chord line (y
        ={chord_y})')
ax.set_aspect('equal')
ax.grid(True)
ax.legend()
ax.set_xlim(min(c1[0], c2[0]) - 4, max(c1[0], c2[0]) + 4)
ax.set_ylim(min(c1[1], c2[1], chord_y) - 4, max(c1[1], c2[1],
        chord_y) + 4)
plt.title('Visualization of Circles and Chord')
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.show()
```